

Group Participation Sheet

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A portion of this course is devoted to group work. This **Group Participation Sheet** (GPS), hereinafter referred to as the **GPS Sheet**, will specifically indicate the group members that actively contributed to the assignment, lab project, deliverable, or report. All names indicated on this GPS sheet will receive the same grade for the attached group work. Group members with names and signatures not shown on this GPS sheet will receive a grade of “**zero**” for the mentioned assessment. Any concerns regarding this matter should be brought to the professor’s attention by the team leader prior to the submission deadline.

Please fill the table below electronically. Group members that participated in this evaluation should **write their names** and append their **signature**. The signed GPS sheet should be added as the **first page of all group evaluations**. This should be **applied** to all evaluations uploaded on SLATE as **softcopy**, and printed/stapled and presented as **hardcopy** to your professor.

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Course Title:	INFO-53883 - AI/ML Capstone Project		
Evaluation Name:	INFO-53883 - D1 (5%, Group Work, Project Proposal)		
Team Number:	INFO-53883 (AI/ML Capstone) - Team 2B		
Referencing and Citation Style:	APA Style ☐	IEEE Style ☒	N/A ☐
Group Members:	First Name and Family Name of group members: (1) Afolabi Adesina (team leader)	Signature of group members: Afolabi Adesina Digitally signed by Afolabi Adesina Date: 2026.05.25 17:28:54 -04'00'	
	(2) Evans Frimpong	Evans Frimpong	
	(3) Xian Qin	Xian Qin	
	(4) Yanan Yang	Yanan Yang Digitally signed by Yanan Yang Date: 2026.05.25 17:26:40 -04'00'	
	(5)		
Submission Date:			

Grade: /5

AI/ML Capstone Project

D1, Project Proposal (5% of final mark)

Due: Week 4 (softcopy on SLATE, hardcopy to your professor in class)

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Instructions:

- Please work on the following deliverable with your group.
- Prepare your deliverable report as discussed in class.
- Type your report using word processing software (preferably using LaTeX, otherwise MS Word is fine).
- Upload your report in PDF format on SLATE. Team leader should upload this deliverable on SLATE as a single PDF file.
- Present a printout (stappled) of your deliverable to your professor in class in Week 4

Q1: Which sector driving modern economies are you considering for your capstone project (i.e., agriculture, education, energy, entertainment, finance, government, health care, insurance, manufacturing, NGO). Select the sector that is mostly relevant to your capstone. Briefly explain your sector selection and what experience you and your group have in this area. Answer this question in one paragraph.

1.1 Industry Chosen: Highway Transportation Safety & Intelligent Systems

The sector selected for this capstone project is highway transportation safety, specifically the application of multimodal Artificial Intelligence (AI) to dynamic risk detection and speed optimization on Ontario's 400-series provincial highway network. This industry sits at the intersection of public safety infrastructure, data science, and real-time decision systems making it a compelling and technically rigorous domain for an applied machine-learning project. Ontario's highway system is among the most heavily used in Canada. The 400-series corridors including Highways 400, 401, 404, and the QEW collectively carry over 500,000 vehicle trips per day during peak periods [12]. Highway 401 alone, stretching 828 km across southern Ontario, is one of the busiest highways in North America and acts as the economic spine connecting Windsor, Toronto, Kingston, and the Quebec border [12].

1.2 Why Our Group Selected This Sector

Our team selected this sector for three overlapping reasons: professional relevance, data accessibility, and social impact.

Professional Relevance: Several group members are pursuing careers in data engineering, intelligent systems, and public-sector analytics. Ontario's Smart Mobility and corridor safety programs are active investment areas, making this domain highly employable.

Data Accessibility: Unlike private corporate datasets, Ontario's highway infrastructure is supported by open-data mandates. The Ontario Data Catalogue, Environment Canada's XML feeds, and the Ontario 511 alert system provide structured, semi-structured, and unstructured data, exactly the multimodal inputs required by our 3-Brain Architecture.

Social Impact: Road fatalities in Ontario averaged over 550 per year between 2018 and 2022, with adverse weather identified as a contributing factor in 25 to 30% of serious collisions [13]. An AI system that can dynamically warn drivers before a hazard materializes rather than after represents a genuine, measurable reduction in preventable deaths.

Q2: Identify a real-world problem that you and your group are interested to explore. Explain the identified "problem" in as much detail that you can. Provide citable data or context to prove that this problem exists and that it matters. Do not forget the famous

saying that: “a problem well stated, is a problem half-solved”. Answer this question in two to three paragraphs.

2.1 The Core Problem: The “Static Data Gap” in Current Navigation Systems

Modern navigation tools (Google Maps, Waze, Apple Maps) excel at solving the “where am I and how fast is traffic moving” problem. They aggregate real-time GPS signals, crowd-sourced incident reports, and historical traffic patterns to recommend routes. However, they share a fundamental and dangerous limitation: they are reactive, not predictive, and they are blind to causal hazards.

This project defines the Static Data Gap as the failure of existing navigation infrastructure to detect and communicate physical road-surface hazards: black ice, wet snow, slush accumulation, sudden whiteouts that are not captured by traffic-flow sensors alone. A road can show zero congestion on Google Maps at 3:00 AM on a February morning on Highway 401, while the pavement surface is coated in black ice invisible to the human eye at posted highway speeds.

2.2 The Scale of the Problem: Data-Backed Evidence

2.2.1 Collision Statistics (Ontario, 2018 and 2022)

According to the Ontario Integrated Collision Database (data.ontario.ca) [11], weather-related factors were identified as contributing causes in highway collisions at the following rates:

Condition	% of Serious Collisions (400-Series Hwy)	Avg. Annual Incidents (est.)
Black ice / road ice	~18%	~310
Snow / blowing snow	~12%	~207
Rain / wet surface	~9%	~155
Reduced visibility	~6%	~103
All weather-related	~45%	~775

Source: Ontario Integrated Collision Data (data.ontario.ca) & Transport Canada Road Safety Annual Report 2023.

These figures reveal that nearly half of all serious highway collisions in Ontario have a weather or road-surface component. This is not a marginal edge case, it is the dominant causal cluster in highway fatalities.

2.2.2 The Economic Cost

The economic impact of weather-related collisions in Canada is estimated at \$10.8 billion CAD annually when accounting for emergency services, property damage, insurance costs,

productivity loss, and long-term care for serious injuries [13]. Ontario, as Canada’s most populated province with the densest highway network, absorbs a disproportionate share of this burden. A conservative 10% reduction in weather-attributable serious collisions on the 400-series network, a plausible outcome of real-time hazard alerting, would represent an estimated \$250 to 400 million CAD in annual cost avoidance and approximately 50 to 80 prevented serious injuries or fatalities per year.

2.2.3 The Technology Gap: What Current Tools Cannot Do

Existing navigation and traffic management technologies face three documented gaps our system is designed to close:

Gap 1, Sensor Blindness: Road-embedded temperature sensors (pavement weather stations) exist but cover only ~4% of Ontario highway kilometres. The remaining 96% of road surface is unmeasured [12].

Gap 2, Latency in Crowd-Sourced Reporting: Waze and similar platforms rely on driver reports. By definition, a report of black ice only exists after the first driver has already encountered it, often after a collision has occurred.

Gap 3, No Cross-Modal Fusion: Environment Canada issues weather warnings. Ontario 511 issues incident alerts. Highway cameras provide visual data. No existing consumer-facing tool fuses all three into a single, real-time, segment-level hazard probability score.

2.3 The Proposed Solution: Ontario Smart-Shield

The Ontario Smart-Shield is a multimodal AI system that addresses the Static Data Gap through three simultaneous data-processing pipelines called the 3-Brain Architecture which fuse textual alerts (NLP Brain), visual road imagery (Vision Brain), and environmental sensor data (Logistic Optimizer) into a single Composite Safety Score (S) ranging from 0 (no hazard) to 100 (extreme hazard). Critically, the system is designed to predict hazard conditions proactively integrating Environment Canada forecast data and incoming 511 alerts before a collision occurs rather than reacting to observed traffic slowdowns after a hazard has already caused harm.

Q3: Identify the prospective clients and stakeholders that the real-world problem that you explained above is targeted for. What are their specific requirements? How are they currently solving this challenge without your AI/ML solution? Support your ideas with reliable market research data. Answer this question in one paragraph.

The target clients for the Ontario “Smart-Shield” include daily commuters on the 400-series highways, commercial trucking fleets, and the Ministry of Transportation Ontario (MTO). Their specific requirement is simple but urgent. They need real-time, hyper-local hazard warnings during bad winter weather. They need to know if a specific road segment is safe to drive on right now.

Currently, drivers rely on standard navigation apps like Google Maps, Apple Maps, or Waze. However, these solutions only optimize for travel time and distance. They track traffic speed, but they suffer from a major “Static Data Gap.” They cannot detect sudden, dynamic environmental hazards like black ice, wet slush, or sudden whiteouts [1]. Because of this gap, drivers are completely blind to changing road conditions. They do not know a road is dangerous until they physically hit an accident zone or a sudden traffic jam.

This is why our AI/ML solution is needed. Market research shows a massive annual spike in multi-vehicle collisions during winter months. This proves that current GPS tools fail to actively protect drivers because their underlying engines treat safety purely as an afterthought to speed [2]. Our model solves this challenge by ranking routes based on actual environmental risk, not just ETA.

Q4: Explicitly justify why an AI/ML solution is needed here rather than applying standard heuristics or traditional software engineering alternatives? What are the benefits of using AI/ML to tackle this real-world problem? Answer this question in one paragraph.

An AI/ML solution is required for this problem because traditional software engineering and heuristics rely on deterministic, rule-based logic that cannot effectively process the high-dimensional, unstructured data streams such as live road imagery and natural language alerts inherent in real-time highway safety. Current industry-standard navigation tools are fundamentally reactive, calculating paths based on observed traffic flow rather than predicting environmental hazards, as their primary architectural focus is limited to the optimization of time and distance metrics [1]. In high-frequency real-world conditions, the volume and velocity of concurrent data streams (dozens of live camera feeds, hundreds of weather stations updating every five minutes, and continuous government alert texts) exceed what any threshold-based rule system can evaluate simultaneously and in real time [5]. Machine learning provides the capability for adaptive, data-driven routing, allowing the system to identify non-linear patterns and correlations between multi-variable environmental inputs such as the specific interaction between wind speed, humidity, and road surface temperature that no fixed heuristic could encode without being manually re-calibrated for every new scenario [8]. Furthermore, deep learning architectures such as Convolutional Neural Networks can extract discriminative visual features from raw pixel arrays to distinguish black ice from wet asphalt, a classification task impossible to encode in deterministic if-else rules while NLP pipelines transform qualitative human observations embedded in government alert text into quantitative hazard indices [5], [3]. The primary benefit of an AI/ML approach is therefore the transition from a reactive observation model to a proactive predictive system, offering a significantly more robust, scalable, and safer solution than static, heuristic-driven alternatives.

Q5: Explain how AI/ML can be integrated into the real-world problem you identified above. Support your ideas from existing literature and/or industry benchmarks. Answer this question in two to three paragraphs.

Integration of AI/ML into the proposed solution follows a multimodal cross-modal fusion strategy. The system applies Natural Language Processing (NLP) to tokenize and clean unstructured text data from sources like official Ontario 511 and OPP emergency alerts, transforming qualitative human observations into a quantitative Textual Risk Index using TF-IDF weighting, a method validated by Gu et al. (2021) who achieved an AUC of over 0.89 for incident detection on government road alert corpora [3]. Simultaneously, a Convolutional Neural Network (CNN) processes live MTO highway camera imagery to classify road surface conditions across four classes (dry, wet, snowy, and icy) producing a continuous Visual Hazard Index that captures texture-level distinctions, such as the reflective sheen of black ice versus the uniform darkening of wet asphalt, which no threshold-based pixel rule can reliably distinguish [4], [6]. These two unstructured-data outputs are then fused with structured environmental telemetry, real-time visibility in km, air temperature, and wind speed from Environment Canada, through a Lasso-regularized Logistic Regression model to produce a single, interpretable Safety Score per road segment. This fusion methodology is directly supported by He et al. (2022), who demonstrated that combining visual, linguistic, and structured sensor indices into a single dynamically weighted composite score outperforms any single-modality model on road hazard classification tasks, with the adaptive weighting mechanism proving especially critical under degraded sensor conditions such as nighttime or dense fog [7].

Industry benchmarks support this integration using standardized, peer-reviewed tools and rigorous hyperparameter optimization. The SAGA solver is selected for the logistic regression model due to its proven convergence efficiency on large, sparse feature matrices under L1 (Lasso) regularization, a property directly applicable to the high-dimensional segment-level feature vectors generated by this pipeline [8]. GridSearchCV or RandomizedSearchCV is then employed to optimize the regularization constant C and penalty type across K-fold cross-validation splits, ensuring the model generalizes unseen highway conditions rather than memorizing training examples [8]. Pan et al. (2019) established that CNN-based road surface classification achieves over 88% accuracy across four surface classes on real highway camera imagery, validating the choice of a convolutional architecture over hand-crafted feature extractors or rule-based pixel thresholds for the visual inference component of this system [6]. By utilizing established open-source libraries, OpenCV for vision preprocessing [4] and NLTK for linguistic structure extraction [3] the system integrates into existing highway data infrastructure as a modular, independently testable plugin, providing a scientifically grounded and maintainable alternative to speed-priority navigation systems [1].

The evaluation framework adopted for this system reflects the asymmetric cost structure inherent to safety-critical ML applications. Rather than optimizing for overall classification accuracy, the primary performance metric is Recall on the HIGH-RISK class, which must exceed 90%, ensuring that no genuine hazard event is missed, a Type II error in this context carries the risk of driver injury or fatality, whereas a Type I error (a false caution alert) causes only minor inconvenience [8]. This evaluation philosophy is consistent with Moosavi et al. (2019), whose analysis of over three million real-world accident records demonstrated that recall-optimized ML models are significantly more appropriate than accuracy-optimized baselines for safety-critical transportation prediction tasks [5]. The broader impact of this integration is a system that transitions highway safety management from a post-hoc, crowd-

sourced slowdown model to a pre-emptive, sensor-driven advisory that can detect hazardous conditions, such as the onset of a lake-effect whiteout on the 401 near London, Ontario, before the first collision is reported, directly addressing a gap that no conventional software engineering approach has the architectural capability to fill.

Q6: The AI/ML capstone project expects groups to work with datasets. In general, groups can train, validate, and test an AI/ML model of their choice. Alternatively, groups may consider customizing an SLM or LLM with specific or proprietary data using techniques like finetuning or RAG. Provide some insights on where will your data come from? Is the data available in open-source, do you need to scrape, generate it on your own, or synthetically simulate it? Provide explanation on licensing, data privacy, and the volume sufficiency needed for training, model adaptation or context injection. Also, is the considered dataset properly labelled or not and what implications will this have on your overall work. Answer this question in three to four paragraphs

Our primary data will be sourced from publicly available government portals, specifically utilizing structured datasets such as the Police Annual Statistical Report - Traffic Collisions [9] and the Traffic Collision Dashboard (2019 to 2022) [10]. These open-source channels provide legally compliant, highly reliable, and authoritative historical data. In terms of licensing and data privacy, government-published open data naturally complies with public sector sharing regulations and comes pre-anonymized, eliminating risks regarding Personally Identifiable Information (PII).

For secondary data, we plan to capture real-time traffic surveillance feeds. To strictly adhere to data privacy laws, we will implement an automated pre-processing pipeline to detect and redact sensitive details, such as human faces and vehicle license plates, before storing or analyzing the imagery. In terms of volume sufficiency, the baseline structured datasets are highly robust, containing approximately 768,000 rows of collision records. This volume is statistically sufficient for training predictive AI/ML models and establishing reliable baseline accident trends.

However, to deepen our analysis, we are currently expanding our scope to ingest external environmental features, including weather patterns, road construction schedules, and local special events. While some of these auxiliary factors will be gathered via structured open APIs, assessing real-time road conditions requires parsing unstructured video streams from traffic cameras. We will build a computer vision pipeline to classify these conditions, supported by a rigorous validation process to ensure the synthesized features are accurate, consistent, and legally cleared for model ingestion.

The dataset presents a hybrid labeling environment, which significantly shapes our project's technical workflow. While the core government collision data is properly labeled with key features (such as time, severity, and location), the unstructured surveillance imagery arrives completely unlabeled. This implies that a substantial portion of our capstone engineering

effort will be dedicated to data labeling and feature engineering. We will need to train a custom computer vision model to identify road hazards and conditions, label those outputs, and then merge them with the structured collision logs. Managing this unlabelled pipeline introduces risks around data alignment and annotation bias, making the validation phase a critical milestone in establishing a credible correlation between environmental hazards and accident probabilities.

Q7: An AI/ML technology stack is the generally composed of the infrastructure layer (i.e., HW/SW), data layer (i.e., injection, cleaning, storage), framework and model layer (i.e., TensorFlow, PyTorch, LLM/SLM, fine-tuning/RAG), ML Ops layer (e.g., Hugging Face, Docker), and the application layer (e.g., APIs, UI, Streamlit). Identify a preliminary technology stack suited to tackle the AI/ML challenge of your capstone project. Answer this question by providing a brief sentence for each AI/ML technology stack.

To construct the “Smart-Shield” predictive routing framework, our technology stack is divided into five specialized layers designed to support our ML architecture efficiently:

Infrastructure Layer: Model prototyping and heavy computational execution will rely on Python 3.10 running on local CUDA-enabled GPUs or cloud-hosted instances to process image and text matrices.

Data Layer: Spatial route mapping, historical accident datasets, and live environmental features will be ingested via the Google Directions API [1], Open-Meteo REST API, and Ontario 511 system.

Framework & Model Layer: The predictive engine utilizes NLTK [3] and Scikit-Learn for text processing (TF-IDF), OpenCV [4] and PyTorch for computer vision (CNN), and Scikit-Learn’s Logistic Regression to calculate the final segment risk scores.

MLOps Layer: To prevent latency during user searches, the ML models will run via asynchronous background pipelines (e.g., automated Cron jobs/Docker containers) to pre-compute risk scores, while hyperparameter tuning is managed by GridSearchCV.

Application Layer: The final product can be deployed as a standalone web application using Streamlit or packaged as a modular plugin/API extension for platforms like Google Maps [1], allowing users to input an origin/destination and receive ranked routes color-coded by dynamic safety scores.

References

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